



C3

Two archetypes, and one company that ran both

~46 min · the sort · what each needs · the Ant Group case · what holds its value

BY THE END OF THIS CAPSULE YOU WILL BE ABLE TO

Place any AI opportunity into one of two archetypes, say what it needs before it is even possible — and defend a strategic recommendation on a real company under real constraints.

Two ways AI enters an industry

Ask one question of any AI opportunity: what existed before? There are only two answers that matter.

ARCHETYPE A — replace a computation you already trusted. Numerical weather prediction ran on supercomputers for decades. GraphCast learned to approximate the answer directly: more accurate on more than 90% of 1,380 tested variables, and a ten-day forecast in under a minute on a single chip instead of hours on hundreds.

What AI supplied: the same answer, for a tiny fraction of the compute. Nothing new is measured.

ARCHETYPE B — ? Before I show you the second one: what is the other thing AI can do to an industry, if it is not making an existing calculation cheaper?

Take a guess: it is not “do it better”. It is something an industry did not have at all.

Takeaway Archetype A is a learned surrogate: cheaper, faster, same question answered.

Two ways AI enters an industry

Ask one question of any AI opportunity: what existed before? There are only two answers that matter.

ARCHETYPE A — replace a computation you already trusted. You have used this without noticing: your maps app's arrival time, a bank's credit scorecard, your phone's night-mode photo. The flagship case is **GraphCast** — the supercomputer's forecast, about three orders of magnitude cheaper, which buys ensembles instead of one best guess.
What AI supplied: a cheaper answer to a question you were already asking.

ARCHETYPE B — manufacture a measurement that did not exist. Also familiar: a watch that scores your sleep and flags an irregular heartbeat; a **credit score for someone who never had a credit file**. The flagship case is **Climate TRACE** — emissions were self-reported and unverifiable, and are now inferred from satellites, thermal signatures and ship transponders across 2,765,771 sources.
What AI supplied: not speed. **Accountability** — a number produced by someone other than the measured party.

Pre-read 4. Third archetype, from Week 1 and much rarer: a capability with no prior at all — AlphaFold. Be suspicious if you think yours is that one.

Takeaway Cheaper answer, or first answer. Almost every use case is one or the other.

What each one needs before it is even possible

The archetype tells you which precondition to check, which risk to plan for, and which tell to watch.

	A • replace a computation	B • manufacture a measurement
Precondition	A long, standardised, labelled history of the old computation's inputs and outputs	An observable proxy, plus a calibration set where truth is known
Existence proof	ERA5 — four decades, standardised, quality-controlled, free. No ERA5, no GraphCast	Combustion is visible from orbit, so emissions can be inferred
What you gain	Speed and cost — which buy volume, frequency and ensembles	Independence — the number no longer comes from the party being measured
How it fails	Inherits the teacher's blind spots; degrades silently outside its training distribution	An estimate gets read as a fact. Uncertainty is published and never survives to the headline
The tell	Nobody kept the old computation's outputs, so there is nothing to learn from	Nothing observable correlates with the thing you want to measure

Most organisations ran the computation for years and threw the outputs away. That is a painful realisation and a cheap fix going forward — start keeping them.

Takeaway Screen an opportunity in thirty seconds: which archetype, and does it have the precondition?

GROUP SORT • 2 MIN

Sort all four. For each one: was this calculation already being done — or is the number new?

1 • A bank replaces its overnight risk simulation with a learned model

2 • An insurer prices roof cover from satellite imagery, with no survey

3 • A plant predicts machine failure from vibration data it already logs

4 • A port replaces its berth-scheduling solver with a learned model

Already being done → A • the number is new → B • 2 minutes in threes, and one of the four is genuinely arguable

The two archetypes across five industries

Sectors are examples inside the buckets, never the structure. A list of applications teaches nothing.

Industry	A • replace a computation	B • manufacture a measurement
Environment	GraphCast / WeatherNext — forecasting at 1/1000 of the compute	Climate TRACE — independent emissions where only self-reporting existed
Finance	MYbank — a loan decision in under a minute, no human — our case	Credit scores for people with no credit file — also our case
Healthcare	Imaging triage, protein folding, trial simulation	Disease signals inferred from speech, gait or retinal images
Manufacturing	Simulation surrogates for design and stress testing	Machine condition inferred from vibration and thermal data
Mobility	Route and network optimisation, learned rather than solved	Road and asset condition inferred from fleet camera data

Row 2 is the same company in BOTH columns, and that is not an error — the strongest cases run both archetypes. The sort is a screen for what to check, not a box to put a company in.

Takeaway Pattern-match by archetype, not by sector. The sector changes; the two questions do not.

All four are archetype B — manufacture a measurement. Which one fails the precondition?

A · Subscriber experience, inferred from network signalling data

B · Diabetic eye disease, graded from a retinal photograph

C · Employee engagement, inferred from meeting and calendar patterns

D · Bridge fatigue, inferred from accelerometer readings already collected

The precondition: an observable proxy, PLUS a calibration set where the truth is known.

What the case actually shows

Three claims. Take them into Week 6, into your project, and into the next strategy deck you are handed.

	The claim	Why it matters beyond Ant
1	Ant ran both archetypes, inside one business. The loan DECISION was A — underwriting a branch officer already did, replaced by a one-minute, zero-human assessment. The SCORE was B — creditworthiness for the ~75% of the population traditional reporting never reached	A made them reachable. B made them assessable. The moat, and the eventual problem, came from B
2	The cost base was the strategy, not a detail. Pennies per transaction is what made a ¥1 product, a 15-cent insurance premium and a one-minute loan possible at all	C2's arithmetic, at national scale. Cost to serve decided which customers existed
3	Removing every internal bottleneck is what created the external one. A firm underwriting six loans a day per officer was never going to be systemically important	The 2020 framework counts only internal bottlenecks — and this one need not be a regulator

Forward to Week 6: governance is not what happens after the strategy. In regulated industries it is a design input, and this case is the argument for saying so.

Takeaway Scale did not attract the regulator by bad luck. Scale is what made one necessary.

The judgement is the product

INSTEAD OF THE GUEST LECTURE

PRE-READS 1-4

When the model is a commodity, everyone will have the same one. So what is left that is worth being good at?

Three things held their value all day. **Selecting** — what a use case buys, and whether the data can feed it. **Pricing** — cost per successful task, the break-even, the sensitivity. **Measuring** — the baseline, the counterfactual, the kill condition. None of these is a model skill, and none of them is going to be automated soon.

Because: the scarce thing was never the capability. It was knowing which capability was worth having.

And one habit worth more than any of them. Separate a claim's **mechanism** from its **calibration**. Mechanisms — the fallback cost decides • a benchmark is a lower bound • scale creates the constraint that stops it — have long half-lives. Calibrations — token prices, model rankings, adoption percentages — go stale in months.

Every number in today's class: will be wrong within a year. Every distinction will not.

This is what the industry guest was going to close on. Everything on this slide is drawn from the four pre-reads — and the habits are the part that outlasts the course.

Takeaway Learn the mechanisms. Re-check the calibrations. Ask who was asked.

CAPSULE 3 • TAKEAWAY

Cheaper answer, or first answer.

Two archetypes, two preconditions, two ways to fail. And one company that got the cost base, the measurement and the scale exactly right — and met a constraint no cost model contains. Next week: that constraint, on purpose.

A2 — what you are building, and when

The group assignment is live. Two minutes on it, because one of these dates is closer than people expect.

What it is. An agent for one of four pre-defined problems — a health-insurance claim, or an outpatient referral — built by your existing course group. **20% of the grade**, instructor-marked, against ILOs 2, 3 and 4.

Eight deliverables: why an agent at all • the agent • the tool layer • guardrails • a 30–50-case eval set • a model battery • **the cost-to-serve model** • two reproduced failures, one of them a loop.

The dates — all 23:59 SGT. **Wed 2 Sep** — the starter scaffold appears on NTULearn. **Fri 4 Sep** — team declaration, a file in the submission folder, not an email. **Sun 13 Sep** — due: the repository, a 2,000-word report, a five-minute recorded demo and the ungraded self-appraisal. **Wed 16 Sep** — peer rating.

Already posted, as of last night: the brief • the FAQ • the self-appraisal template • and **reference data for both problems — 15 labelled cases each**, with a checker script and a data dictionary. Only the starter scaffold is still to come.

D6, the cost-to-serve model, is C2's three-layer method on your own problem's volumes. Teams are the existing course groups, not re-formed. A missing self-appraisal counts as an incomplete submission.

Takeaway Everything except the scaffold is already in your hands. The date that bites is Friday.

Coming up — Class 6

The final session. Five weeks building the thing; one week on what it does when it goes wrong.

	What we cover	What you will do
1	Failure modes, per kind of AI	Hallucination, agent loops, ML bias, embodied-AI safety — the silent-failure thread from Week 2, closed
2	Responsible deployment	Eval, observability, guardrails — plus the OWASP LLM Top 10 as a map of the attack surface
3	The regulatory landscape	Singapore IMDA • India DPDP • China CAC • EU AI Act • US NIST — mapped to your home markets
4	Red-team your own build	Attack the agent you built, then write the guardrail that stops the attack
5	The frontier, and the last case	Reasoning models, multimodal, embodied AI — and Governing OpenAI (A) and (B)

Takeaway A2 is due that day, and project pitches run in the session. Details firm up nearer the date.